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Learning Personalized Alignment for Evaluating Open-ended Text Generation

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Which one do you prefer?

🥳 Life is hard, let’s read something happy!

Bad endings are more impressive

Premise: A successful artist Samantha struggles with the emotional aftermath.

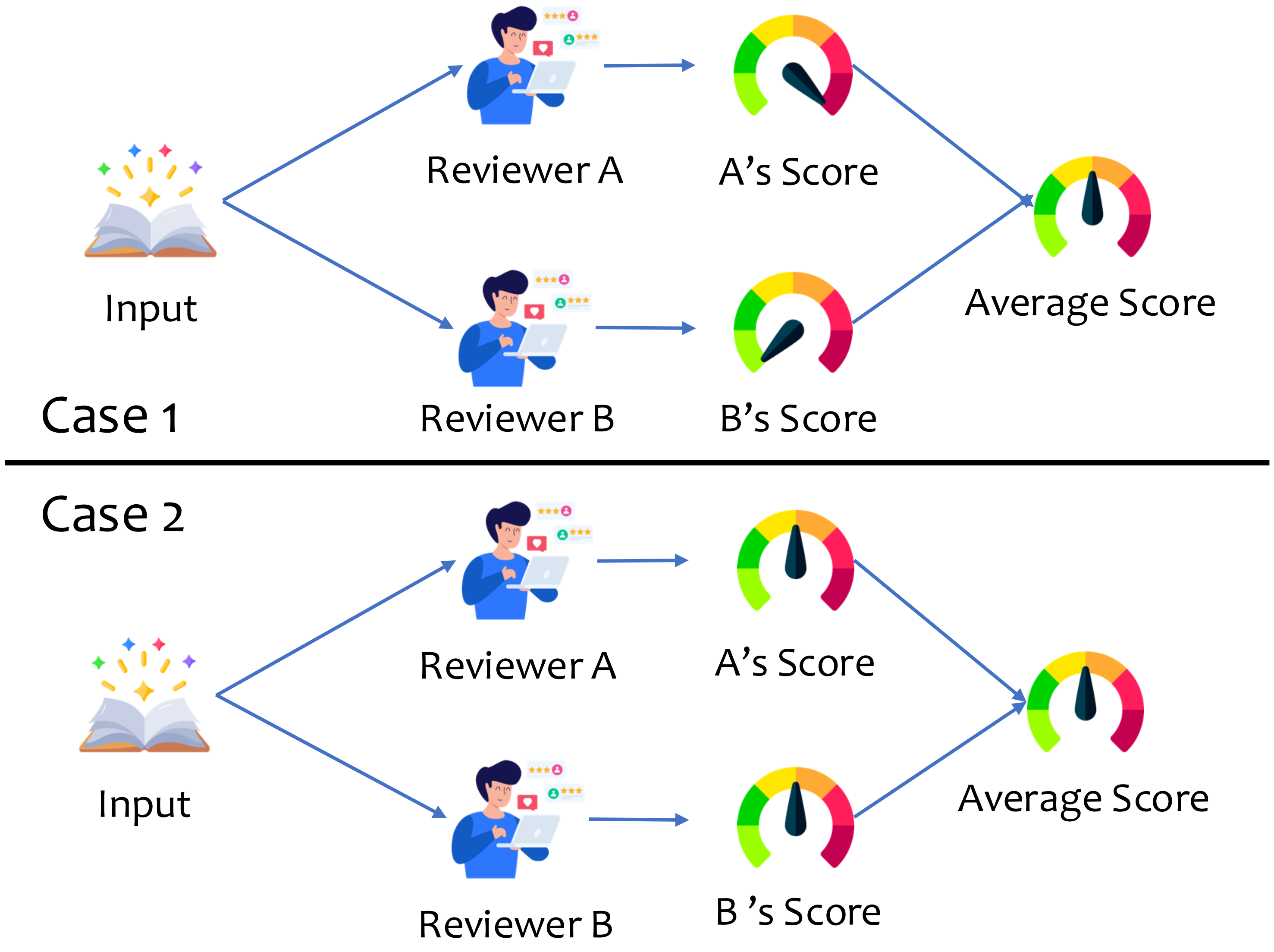
Plot A:
… Samantha hesitantly agrees to the idea and begins working on the new series, finding solace in the creative process.

Plot B:
… Ben (Samantha’s son) becomes increasingly worried for his mother and reaches out to Jesse and Isabelle (Samantha’s friends) for support **but fails.**

I like **Plot A** because it has an uplifting ending. Plot B is sadder. There isn't a happy resolution to the story.

I found **Plot B** more interesting because it involves a family dynamic, which adds a layer of complexity to the story. Plot B also explores how the characters are dealing with emotional pain and loss, making it more relatable and empathetic.

One-score-fit-all verse Personalized Score

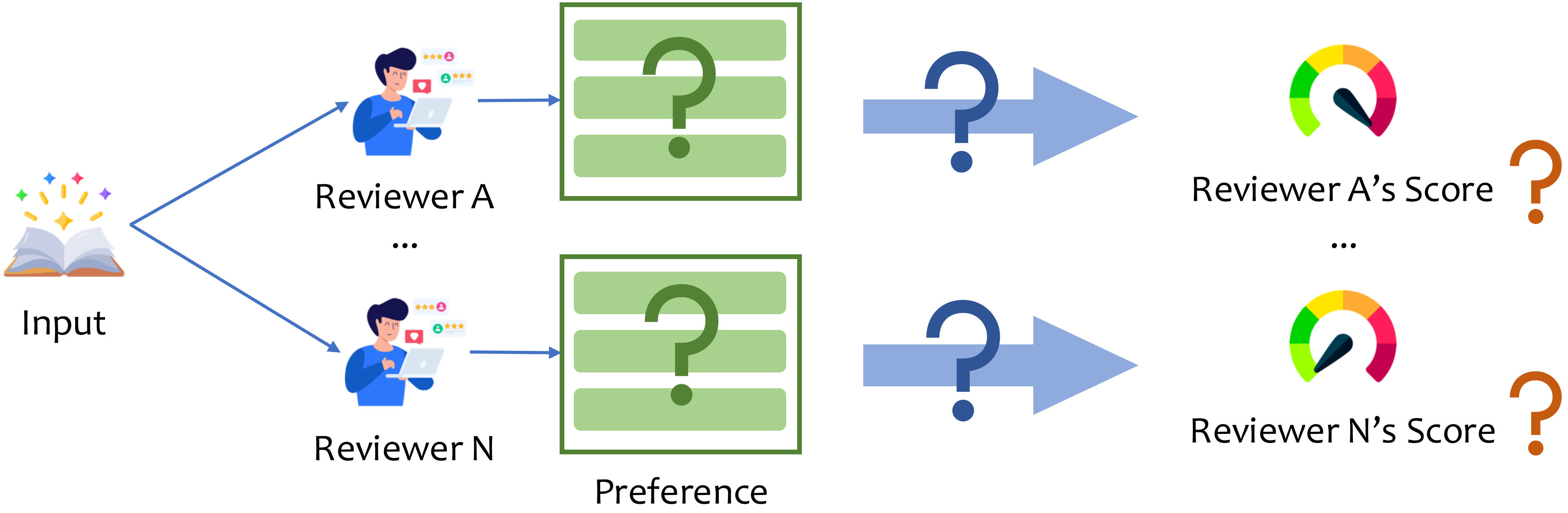


Are these two cases the same?

For open-ended generation, we usually get case 1
-> low inner-annotator agreement

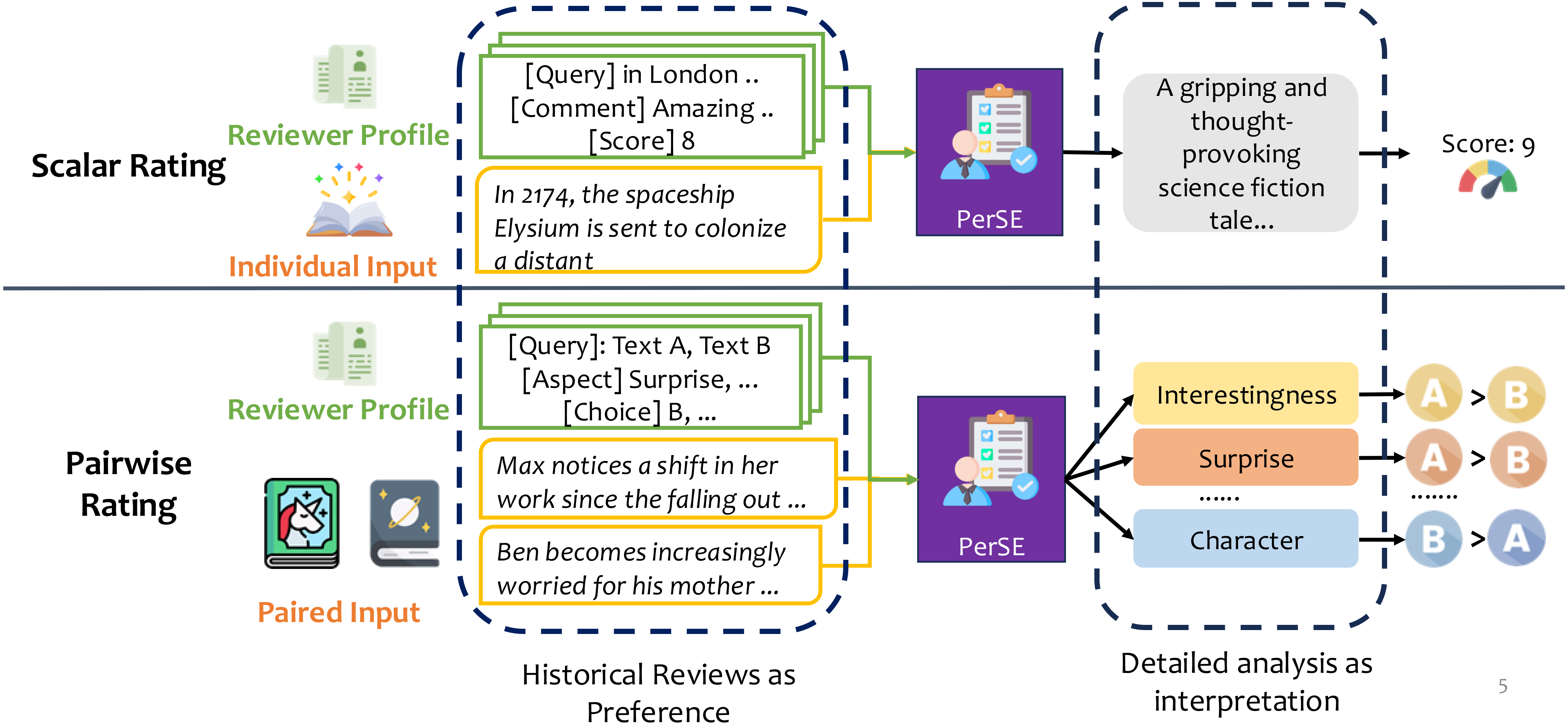
What if we want to tailor generation for personalization?

Learning Personal Preference



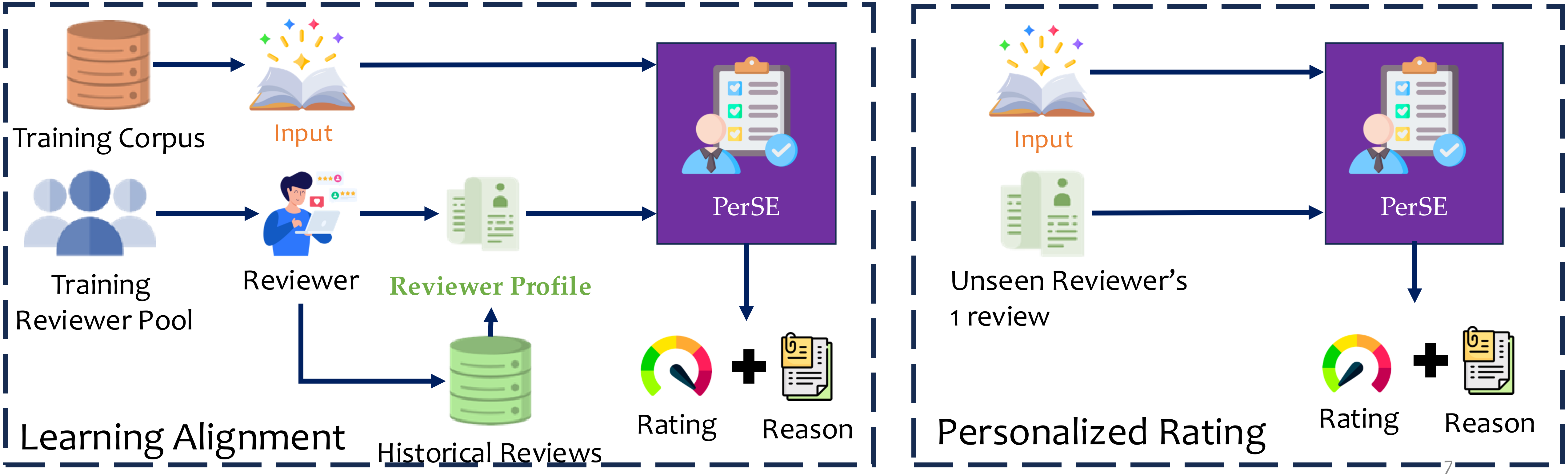
How to collect personal preference from reviewer?
How to infer personalized score from preference?
How to interpret the personalized score?

PerSonalized Evaluation Model (PerSE)



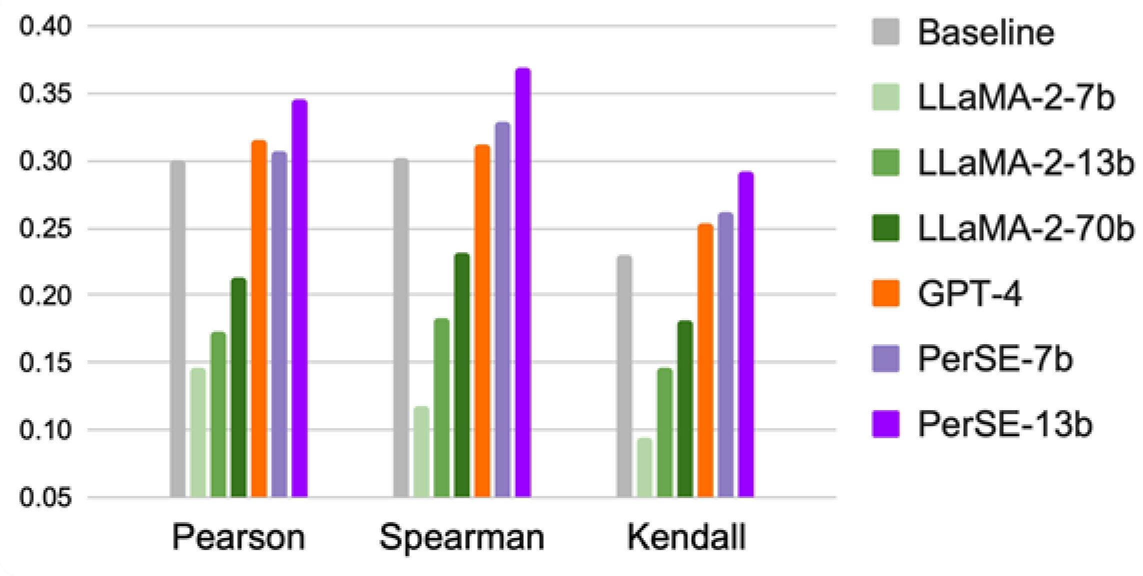
Learning Personalized Alignment in Evaluation

Enhance LLM's capability in aligning with specific preference with limited personal context and detailed explanation

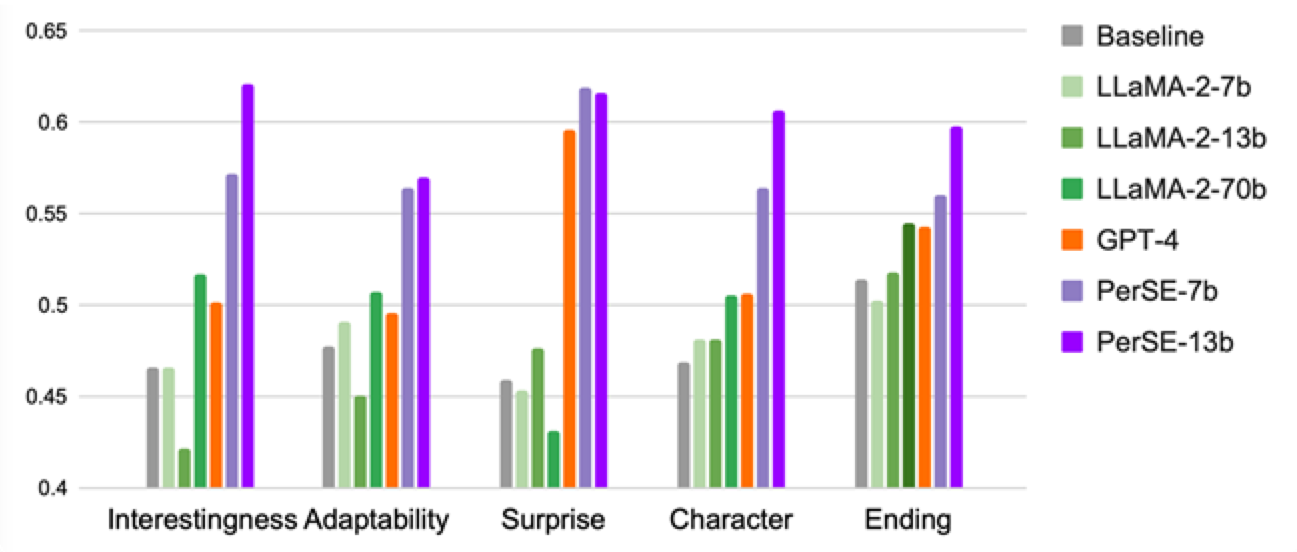


PerSE significantly outperforms all baselines

Highest Correlation with Human Rating on Scalar Rating

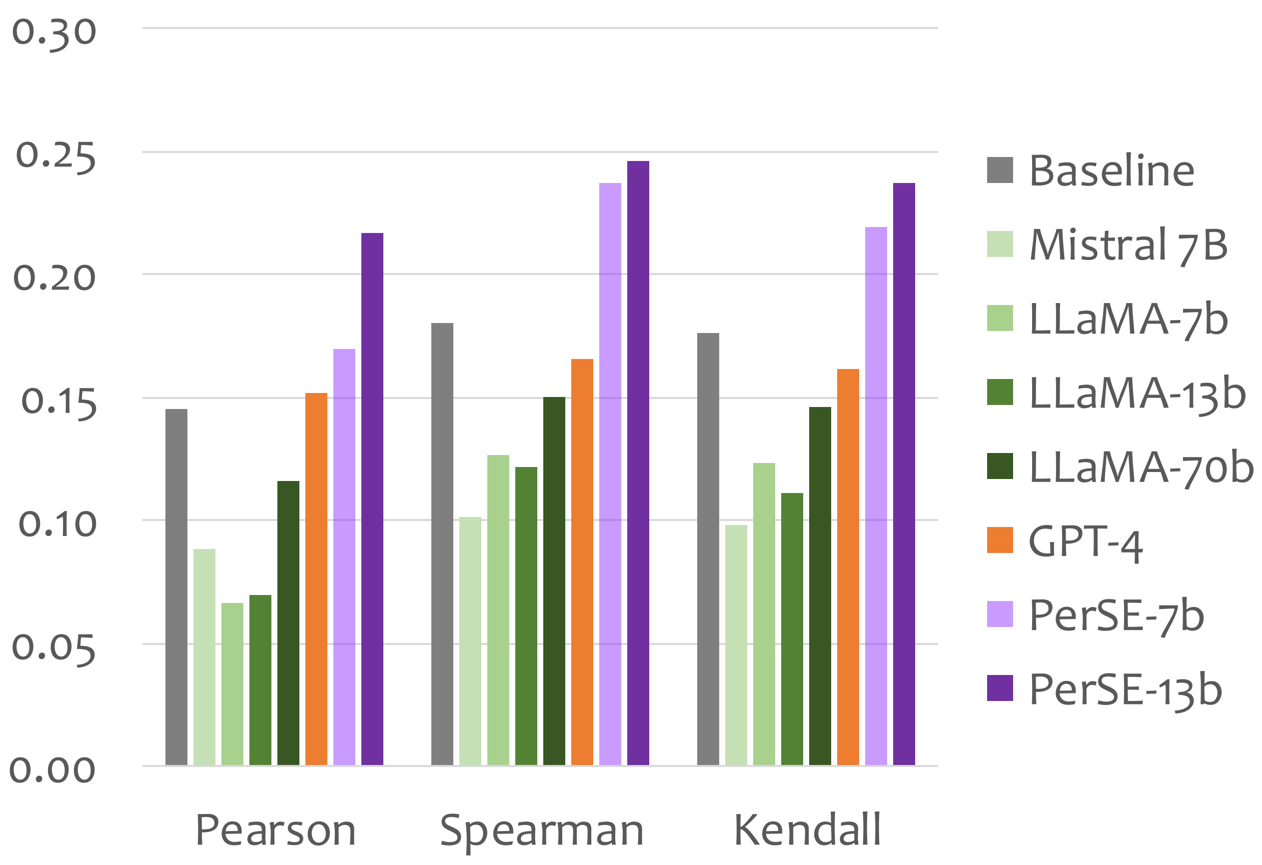


Best accuracy for Pairwise Rating on different aspects



PerSE Generalizes Well to Other Domains without Fine-tuning

Directly Apply PerSE trained on Per-MPST to Amazon Book dataset



PerSE cares more about reviewer's preference, while GPT-4 is more general

Reviewer Preference

A newscast is shown about a year after a prehistoric piranha attack on Lake Victoria. The lake is now uninhabitable, and the town has been abandoned due to the drying up of the m

[R ... tures ev ... l , a ba

Ev ... girl wl ha

[R ... nt you just hate it when this happens? Now that you finally found yourself a gorgeous looking high school sweetheart, she turns out to be a murderous witch, capable of making you eat glass beer bottles with the touch of her hand! **[Score]** 5

critical about the plots, and especially cares about the novelty and vision

Plots

Professor Jack Hanson and his colleagues go to the Gibraltar ice shelf to take ice core samples anc ... entation to the UN that is poorly rec ... th Atlantic show a sharp drop in oce ... believes Jack's theories and works with him to build a forecast model....

Big topic but boring

Reviewer

I mean, you ... The whole thing may b ... appealing to the audie ... e old, disastrous ... gain.
[Score] 2

Not appealing, repeated
Score: 2

GPT-4

This film offers a chilling, literally and metaphorically, journey into the devastating effe ... t's often brushed aside... ... ear of the unknown future of our planet to craft an engaging narrative. **[Score]** 7

Good topic Score: 7

PerSE

[Review] I r ... the movies are. ... tch, although ev ... s. Too bad they di ... a decent mov

visible budget restrictions
score: 3

Take Away

- LLMs finetuned with human feedback is general
  - Learn common interest but struggle with diverse preferences
- PerSE: instruction tune on a small amount of preference data
  - help LLMs better align with specific preferences

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